A report on: Feature Importance-aware Transferable Adversarial Attack

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I. Abstract

A key aspect of any adversarial attack is to transfer the attack on any unknown model so that it can easily be applied without the issue of overfitting to the source model it is crafted on. The existing attacks focus on distorting arbitrary features of the object in an image to have an impact on the model's prediction accuracy. Such indiscriminate feature distortion makes the adversarial attacks get trapped into local optimum. The authors of the paper advocate a special method with which the transferability can be achieved by opting to only distort the important features of the object in a clean image. This is done by a method called the gradient aggregation which takes the averages of the gradients with respect to the feature maps computed on a batch of random transforms of the original image. Such calculated gradients show invariance across different models and are also highly correlated with the object in target. Such attacks make sure that the intrinsic features of objects stay intact while at the same time the model-specific information are suppressed.

II. Paper summary

A. Introduction

In recent times, deep neural networks (DNN) play many crucial roles in vision tasks. But it has been observed that the DNNs can easily be misguided by adversarial attacks to predict incorrectly. It raises safety concerns in many real-life application fields such as autonomous driving. On the other hand such attacks also point out the drawbacks in the model and helps in improving it's robustness. Among different adversarial attacks, the transfer based black box attacks are more popular due to their practicality and flexibility. In order to enhance the transferability, the intermediate layers of the DNN are targeted for attack instead of the outer layer. But such attacks distort the features in the intermediate levels without becoming aware of the important features of the object in an image, this leads to overfitting issues. The main reason behind this is the learning of the noisy features of the source model by the adversarial attacks.

This paper introduces a Feature Importance-aware Attack (FIA) which eliminates the issue of overfitting and helps with the transferability of the adversarial attacks to other models. The gradient aggregation misleads the model predictions. Random transformation (random pixel dropping for example) are first applied to the original image. Such transformation preserves the spatial structure and texture but the non-semantic details vary. The object specific features will be the same while the rest of the features in the image are changed into noise. The gradients are computed based on the transformed features which in turn generate the transferable adversarial attacks.

In this report, firstly the method of Feature Importance-aware Attack (FIA) is discussed. Secondly, the experimental results are presented which compares the performance of FIA transferable attacks against other state-of-the-art transferable attacks. Lastly, the attack in discussion is comapred with it's counterparts followed by summary.

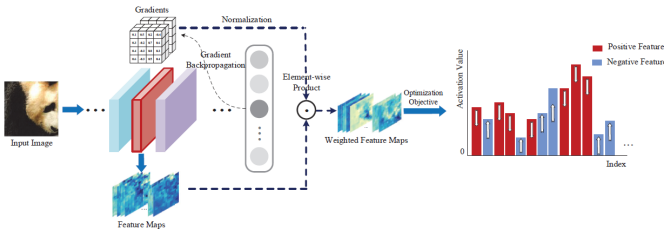


Fig. 1. Overview of Feature Importance-aware Attack (FIA).

B. Fundamentals of generating the adversarial attacks

An arbitrary neural network is given by $f_\theta : x \mapsto y$ with x being the original image and y being the true label. The input x is modified by adding a small perturbation ϵ which is $x^{adv} = x + \epsilon$ so when x^{adv} is fed to the model, the predicted label doesn't match with the true label i.e $f_\theta\ (x^{adv}) \neq y$. ℓ_p-norm is typically used to regularize the perturbation. The below equation (1) is optimized to generate the adversarial examples

$$\arg\max_{x^{adv}} J(x^{adv}, y)\ \text{, s.t. } \|x - x^{adv}\|_p \leq \epsilon, \qquad (1)$$

where $J(\cdot,\cdot)$ is the loss function measuring the distance between the actual label and the predicted label and $p = \infty$. In case of black box attacks, the attacks are crafted on f_ϕ where the parameter set ϕ is accessible. After that the example is transferred to attack an unseen model f_θ where θ is not available to the attacker.

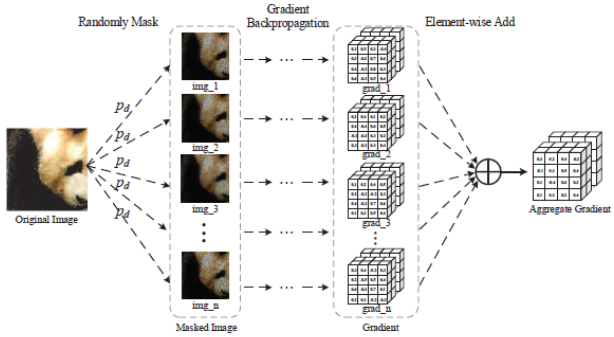


Fig. 2. Illustration of the aggregate gradients.

C. Feature Importance

The basic intuition to make the model misclassify an image is to distort the important features of the object in the image. But the classifiers have a tendency to fit to the data domain where it learns some noisy features along with the important object-aware ones to improve the model accuracy. The indiscriminate distortion of features without the knowledge of such model-specific features traps the adversarial attacks to overfit to the source model on which it is being generated. As a result, the attack performs poorly on an unseen model when being transferred. The key idea for enhancing the transferability of the attacks is to focus on distorting the model-agnostic critical features (for example basic semantic concepts of an object) on the source model which is referred to as feature importance. This is achieved through a method called aggregate gradients.

D. Aggregate Gradient

Let f be the source model and $f_k(x)$ denotes the feature maps of the k-th layer in the model. The feature maps are generated by applying specific filters on the input image. The gradients of the outputs of the model w.r.t $f_k(x)$ is considered as the feature importance contributes to the model decisions.

$$\Delta_k^x = \frac{\partial l(x,t)}{\partial f_k(x)}, \qquad (2)$$

$l(\cdot,\cdot)$ in equation (2) denotes the logit output of the model. Δ_k^x is the raw gradient which includes all the model-specific information. Aggregate gradients help suppressing the model-specific information. Random pixel dropping (or randomly mask) with certain probability p_d is applied to x as shown in figure 2, then the gradients are calculated each time and summed up at the end. The aggregate gradients are thus represented as the following,

$$\bar{\Delta}_k^x = \frac{1}{C}\sum_{n=1}^{N} \Delta_k^{x \odot M_{p_d}^n}, M_{p_d}^n \sim \text{Bernoulli}(1 - p_d), \qquad (3)$$

where M_{p_d} is the binary matrix with same size as x and $\odot$ denotes the element-wise product. C is the normalizer obtained through ℓ_2-norm on the corresponding summation term. N is the number of times the random masking is applied on x. Model-agnostic (trivial) features are not changed by such transformation but the model-specific features are neutralized due to the vulnerability to such transformation. The overview of the FIA transferable attacks can be seen in figure1.

E. New optimization

The new loss function is now defined as

$$\mathcal{L}(x^{adv}) = \sum(\bar{\Delta}_k^x \odot f_k(x^{adv})), \qquad (4)$$

where $\bar{\Delta}_k^x$ is the aggregate gradients calculated from the feature maps and $f_k(x^{adv})$ is the tweaked k-th

layer feature maps. The gradients show the change in the intensity of colour at a certain pixel between a modified and an original image. The aggregate gradients of the important features are given higher intensity. We can get back the true label by increasing or decreasing the intensity of the aggregate gradients. So the positive $\bar{\Delta}_k^x$ is decreased and the negative $\bar{\Delta}_k^x$ are increased which is achieved through minimizing the equation (4). The $J(\cdot,\cdot)$ from equation (1) is substituted by $\mathcal{L}(x^{adv})$ in equation (4) to generate the feature importance-aware transferable adversarial attack as below

$$\arg\min_{x^{adv}} \mathcal{L}(x^{adv}) \text{ , s.t. } \|x - x^{adv}\|_\infty \leq \epsilon, \quad (5)$$

F. Experiment results

The proposed FIA is tested and evaluated on twelve state-of-the-art classification models in which seven are normally trained models and five are defense model (adversarially trained model). The input dataset consists of 1000 images. The effectiveness of the proposed FIA are checked as well against those of different state-of-the-art adversarial attacks. The parameters are set as follows, the perturbation $\epsilon = 16$, the drop probability $p_d = 0,3$ in case of the normally trained models and $p_d = 0,1$ for the defense models and the ensemble is $N = 30$.

	Attack	Inc-v3	Inc-v4	IncRes-v2	Res-50	Res-152	Vgg-16	Vgg-19
Vgg-16	MIM	80.3%	81.1%	74.6%	89.3%	84.4%	**100.0%***	96.5%
	DIM	87.2%	87.0%	80.9%	92.0%	87.8%	99.8%*	98.9%
	PIM	84.1%	82.0%	75.6%	91.1%	85.9%	**100.0%***	98.9%
	PIDIM	89.1%	89.5%	84.7%	93.8%	90.8%	99.9%*	98.8%
	NRDM	73.6%	72.8%	57.1%	77.5%	73.0%	93.2%*	91.1%
	FDA	76.1%	76.7%	64.0%	81.7%	78.7%	95.7%*	95.7%
	FIA	95.7%	95.6%	92.3%	97.3%	95.3%	99.8%*	99.6%
	FIA+PIDIM	**97.6%**	**97.5%**	**93.8%**	**98.2%**	**96.4%**	99.8%*	**99.8%**

Fig. 3. Success rates of different attacks on normally trained models. The first column is the source model (only $VGG-16$ is shown for better readability). The first row shows the target models. '*' indicates the white box attacks since the source model and the target model are the same. The best results are highlighted in bold[1]

1) Attacking Normally Trained Models: As seen from the figure3, FIA achieves 90% success rates against each of the normally trained models with the source model $VGG-16$ whereas other transferable attacks achieve merely 60%. Furthermore it has been seen that FIA can easily be combined with other transferable attacks to improve the performances.

	Attack	Adv-Inc-v3	Adv-IncRes-v2	Ens3-Inc-v3	Ens4-Inc-v3	Ens-IncRes-v2
Vgg-16	MIM	64.3%	61.1%	64.3%	64.3%	45.0%
	DIM	69.9%	66.2%	70.3%	67.8%	49.9%
	TIM	52.8%	46.2%	55.1%	55.3%	41.6%
	PIM	51.9%	43.2%	50.2%	56.3%	39.9%
	TIDIM	59.1%	48.2%	59.6%	60.3%	47.9%
	PITIDIM	51.0%	44.6%	55.6%	60.7%	43.1%
	FIA	**87.8%**	**86.3%**	**85.6%**	**86.0%**	**70.8%**
	FIA+PITIDIM	74.7%	71.4%	77.3%	80.1%	67.0%

Fig. 4. Success rates of different attacks on defense models. The first column is the source model (only $VGG-16$ is shown for better readability). The first row shows the target models. '*' indicates the white box attacks since the source model and the target model are the same. The best results are highlighted in bold[2]

Finally, FIA is also compared alongside the performance of other feature-level attacks such as NRDM and FDA where FIA proves to be more effective. It has been seen that the simple models yield high transferable adversarial attacks as compared to complex models.

Attack	Adv-Inc-v3	Adv-IncRes-v2	Ens3-Inc-v3	Ens4-Inc-v3	Ens-IncRes-v2
MIM	71.0%	69.6%	70.6%	71.3%	51.7%
DIM	81.2%	83.6%	81.6%	79.8%	65.9%
TIM	68.0%	63.9%	70.7%	72.7%	60.7%
PIM	72.1%	66.4%	76.3%	79.3%	68.2%
TIDIM	74.4%	68.3%	75.9%	77.8%	67.1%
PITIDIM	72.1%	66.4%	76.3%	79.3%	68.2%
FIA	**90.9%**	**90.0%**	**88.0%**	**88.4%**	**75.8%**

Fig. 5. Success rates of different attacks on defense models while crafted on ensemble models $Res-50, Res-152, Vgg-16, and Vgg-19$.

2) Attacking Defense Models: Defense models exhibit that they can protect themselves better against the attacks in figure4. Hence the performance percentages drop. Other feature based attacks perform quite badly as compared to proposed FIA. The combined FIA+PITIDIM outperforms every other attacks in this case. Ensemble models are used as well to generate the adversarial attacks to improve the transferability as the attacks show lower transferability in the defense models shown in figure5.

3) Effects of Parameters in FIA transferable attacks: High drop probability p_d such as 0.5 makes the success rate significantly lower as it destroys the structural information of an object. The p_d is optimal

between 0.2 and 0.3 for normally trained models and around 0.1 for the defense models. Thus the p_d is optimally kept at 0.3. The optimal ensemble number N is 30 because after this point the effects of N seem to become gradually stable. The middle layers have been selected to distort the features to generate the adversarial attacks. The early layers make it difficult to distort the features as they lack the semantic concepts of true classes whereas the further layers contribute heavily to improve the model accuracy thus the further layers are model-specific. Hence the selection of middle layers to generate the transferable attacks is the right choice.

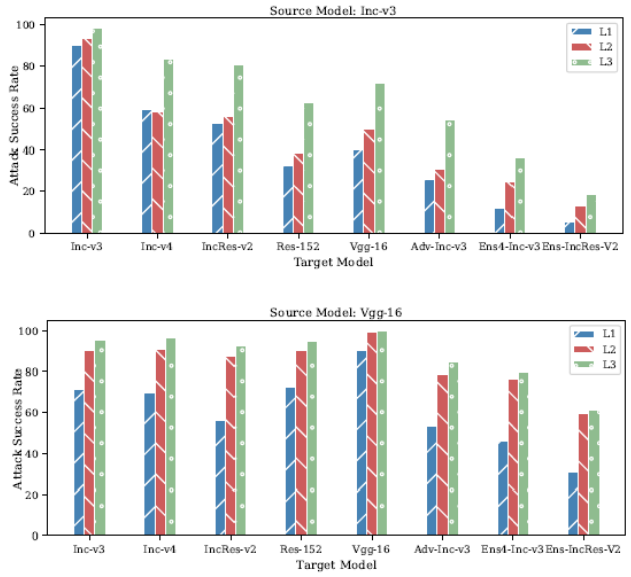


Fig. 6. Effect of aggregate gradient on attack success rate.

4) Effects of the Aggregate Gradient: The aggregate gradients make the adversarial attacks the most transferable across the target models. Three of the below objective functions are generated as part of an ablation study to understand the important contribution of the aggregate gradients to the transferability factor

$$\begin{aligned} \mathcal{L}_1 &= \sum \mid f_k(x) - f_k(x^{adv}) \mid, \\ \mathcal{L}_2 &= \sum (\Delta_{clean} \odot (f_k(x) - f_k(x^{adv}))), \\ \mathcal{L}_3 &= \sum (\Delta \odot (f_k(x) - f_k(x^{adv}))) \end{aligned} \tag{6}$$

where $\mathcal{L}_1$ is the feature divergence without constraint, $\mathcal{L}_2$ uses the gradient from the original clean image and the $\mathcal{L}_3$ is our proposed loss. Nonetheless $\mathcal{L}_3$ outperforms the other two loss functions indicating it's high effectiveness as depicted in figure6.

III. Other transferable adversarial examples

1) Adversarial patch: Adversarial patch aims to add an image-independent patch to the model input which is highly visible to human eye. So such approach doesn't try to modify the input images by subtly adding small perturbations. As these patches are not dependent on the input image, the attackers don't need the knowledge of image information to implement the attacks. Adversarial patch makes itself a transferable attack in a way where once the attack is generated it can be shared widely over the internet for other attackers to print it out and apply to other target models. Also the existing defense technique against the small perturbation become useless to such large perturbation attacks.

The examples are generated as the following. First, the patch is masked so that it can take any shape, then the masked patch is trained on a number of images while applying random translation, scaling and rotation on the patch in each image. Lastly, the gradient descent is used to optimize it. Thus such perturbation is claimed to become universal as it can be applied on any background. [2]

2) Universal adversarial perturbations - UAPs: UAPs generates the adversarial examples using a single small perturbation vector which fools the models to misclassify the targets with high probability. These small quasi-imperceptible perturbations don't significantly affect the underlying data distributions as they are difficult to detect. Hence the data-dependent optimizations are avoided while computing perturbation to a new data point. So the mere addition of the universal perturbation to the image is enough to make the model produce incorrect results.

Let μ denote the distribution of the input image x in R^d. $\hat{k}$ defines the classification function which gives an estimated label $\hat{k}(x)$ for each image x. A perturbation vector $v \in R^d$ is added to the input x to fool the model such that

$$\hat{k}(x+v) \neq \hat{k}(x) \text{ for "most" } x \sim \mu \tag{7}$$

So v is the image-agnostic perturbation for which the estimated label differs from the ground truth

for most of the data samples coming from the distribution μ. It has also been shown that UAPs not only generalize well on new data points in natural images, it also well generalizes on different model architecture thus showing strong transferability of the UAPs attacks. [5]

3) Carlini-Wagner (CW) attack: The Carlini-Wagner attack is a powerful optimization-based attack that generates the adversarial examples. Three distance based metrics L_0, L_2 and L_∞ are used to generate the attacks. L_0 measures the number of pixels being altered in an image. L_2 is the standard Euclidean distance measure between two pixel points. It can remain small when there are many changes to many pixels. L_∞ measures the maximum change to any coordinates. Through this approach, the adversarial examples are generated as below

$$\begin{aligned} &\text{minimize } \mathcal{D}(x, x+\delta) \\ &\text{such that } \mathcal{C}(x+\delta) = t \\ &\qquad x+\delta \in [0,1]^n \end{aligned} \tag{8}$$

where x is the input and fixed. $\mathcal{D}$ is the distance metric. The main aim is to find the δ which minimizes $\mathcal{D}(x, x+\delta)$. Gradient based optimization approach is used to minimize the distances. So that the transformed input and original input stay almost similar and the misclassification of the model is maximized. t is a target class selected by the attacker. [3]

4) Boundary attack: The decision-based boundary attack is more practical to real-world application as it is independent of the model gradient information, it also doesn't depend on the confidence score and unlike the transfer-based attacks the boundary attack depend very less on the model architecture or the training data. It is simpler to apply as the boundary attack solely depends on the decision of the model. To generate the attack the adversary has to observe the final decision of the model while knowing at least one perturbation. So the boundary attack requires a sample which is already adversarial. So the attack generation starts with large perturbation and gradually reduces the perturbation while still being on the adversary boundary. An input is considered as adversarial when the model assigns some class to the perturbted input which is different from the class label of the original input. Boundary attack is very flexible to generate as long as there exists an initial adversarial input. There is close to no hyperparameter tuning: the length of the total perturbation and the length of the step towards the original input.[1]

5) Momentum Iterative Fast Gradient Sign Method (MI-FGSM): Usually the adversarial attacks generated by the iterative methods exhibit weak transferability. MI-FGSM eliminates the issue of weak transferability. In the Fast Gradient Sign Method (FGSM) the adversarial attacks are generated by adding the sign of the gradients to the input which maximizes the loss function. Iterative FGSM (I-FGSM) applies the FGSM multiple times with a small step size to meet the L_∞ bound. The MI-FGSM includes a velocity vector in the gradient direction of the loss function through iterations to make the update direction stable and escape the local maxima. The velocity vector is accumulated by a method called the momentum method which helps accelerate the gradient descent across iterations. The momentum method is also very effective in Stochastic Gradient Descent (SGD) to stable the updates. The previous gradients are memorized by the method to avoid trapping into local minima and maxima. [4]

6) Comparison with the FIA transferable attacks:

a) Adversarial patch: Even though the adversarial patch works for any background. The patch needs to be applied on a specific location in the image making it target model specific. This reduces the transferability of the attacks to other models. Also the adversarial patch fails to attack various models where the input image has undergone different transformation such as resizing, changed angle, different lighting conditions and so on. On the other hand, the proposed FIA targets distoring the important features of the object in an image which are almost common in various models. Thus the FIA proves to be more transferable than the adversarial patch attacks.

b) Universal adversarial perturbations - UAPs: As we have seen that the proposed FIA focuses on the important model agnostic features to maximize the misclassification of the model. So FIA attacks generated on one model can be transferred to other target models to attack when

the target model also share the same important feature sets. But if the target model architecture uses complex and distributed representations, then FIA may not be well transferred like UAPs.

c) Carlini-Wagner (CW) attack: One key aspect of generating the CW attack is the use of the low perturbation which is difficult to detect. It has been shown that such attack can be transferred well to other models. CW attacks performs nicely on both the white box and the black box attacks. But the CW attacks can become model-specific due to it's optimization of the specific loss function. Also it is difficult to generate the CW attacks if any new data is far from the training data distribution. As FIA aims to distort the features which are important basic feature sets across models, it may perform well on specific model architectures which share the same important semantic concepts.

d) Boundary attack: Even though the boundary attacks seem to be conceptually simple and has shown to transfer to other models well, it is computationally expensive.

e) Momentum Iterative Fast Gradient Sign Method (MI-FGSM): MI-FGSM are shown to be highly effective in transferring the adversarial attacks to other models by smoothing out the perturbations. It also achieves high misclassification error rates at the same time. But MI-FSGM is highly computationally expensive as a large number of queries are required to generate the adversarial examples. As a result FIA attacks are preferable sometimes as it involves only identifying the important features of an object and adding perturbation to those features to make the misclassification.

IV. Summary

The weak transferability of the adversarial attacks to other unseen target models has been a major problem in the existing black box attacks. The proposed FIA transferable attacks overcome this issue by improving the transferability of the examples. It generates the adversarial attacks by mindfully distorting only the trivial features (called the important features in the paper) in the intermediate layers and calculates the aggregate gradients at the end. The model predictions are misguided by decreasing the non-trivial features and increasing the trivial features. Though the authors of the proposed method suggest that the aggregate gradients ensure the most transferability of the adversarial attacks but in practice the transferability of the attacks depend on a number of factors such as the model architectures being used, different training datasets, the defense mechanism against such attacks used by the target models, specific applications and so on. The Feature Importance-aware (FIA) seems to transfer well to those target models which also depend on specific features as the source models the FIA transferable attacks are crafted on. Still this approach is relatively new and needs to be holistically studied for accepting it's universally transferable claim.

References

[1] Wieland Brendel, Jonas Rauber, and Matthias Bethge. Decision-based adversarial attacks: Reliable attacks against black-box machine learning models. 2017.

[2] Tom B Brown, Dandelion Mané, Aurko Roy, Martín Abadi, and Justin Gilmer. Adversarial patch. 2017.

[3] Nicholas Carlini and David Wagner. Towards evaluating the robustness of neural networks. In *2017 ieee symposium on security and privacy (sp)*, 2017.

[4] Yinpeng Dong, Fangzhou Liao, Tianyu Pang, Xiaolin Hu, and Jun Zhu. Discovering adversarial examples with momentum. 2017.

[5] Seyed-Mohsen Moosavi-Dezfooli, Alhussein Fawzi, Omar Fawzi, and Pascal Frossard. Universal adversarial perturbations. 2017.